Escuela de TGG, Oaxaca.

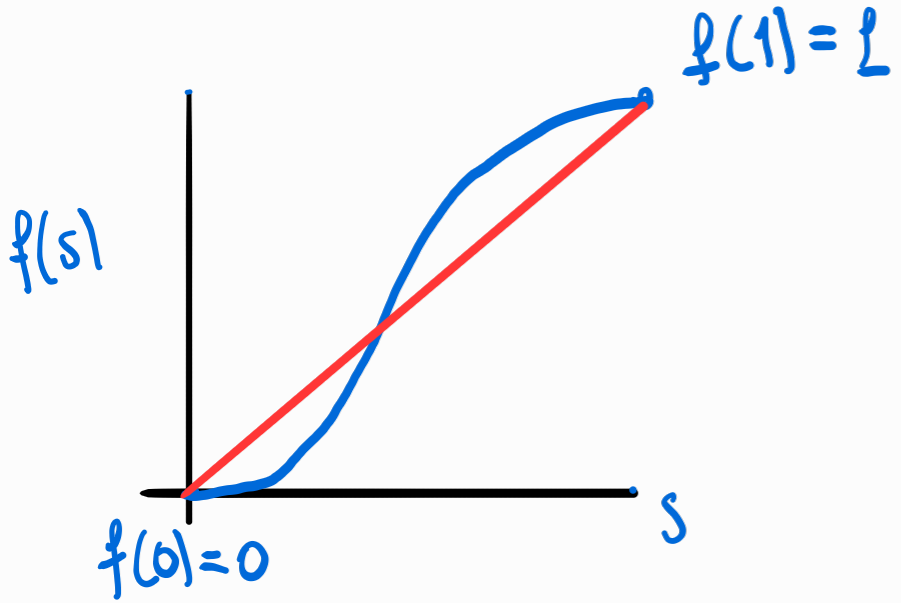
El Grupo de Thomson

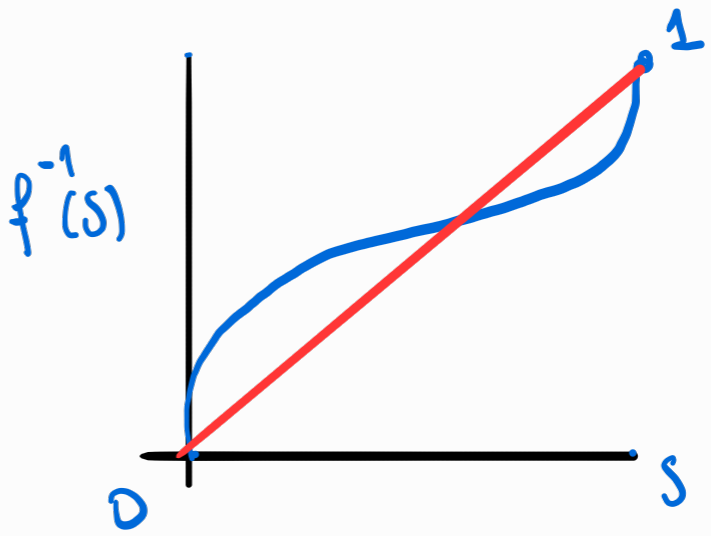
- Juan Parra Flores

Curso: Francesco Fournier Facio

$$\text{Homeo}_+[0,1] = \left\{ f:[0,1]\to[0,1] \;\middle|\; \begin{array}{l}\text{creciente}\\ f(0)=0 \quad f(1)=1\end{array} \right\}$$

Grupo

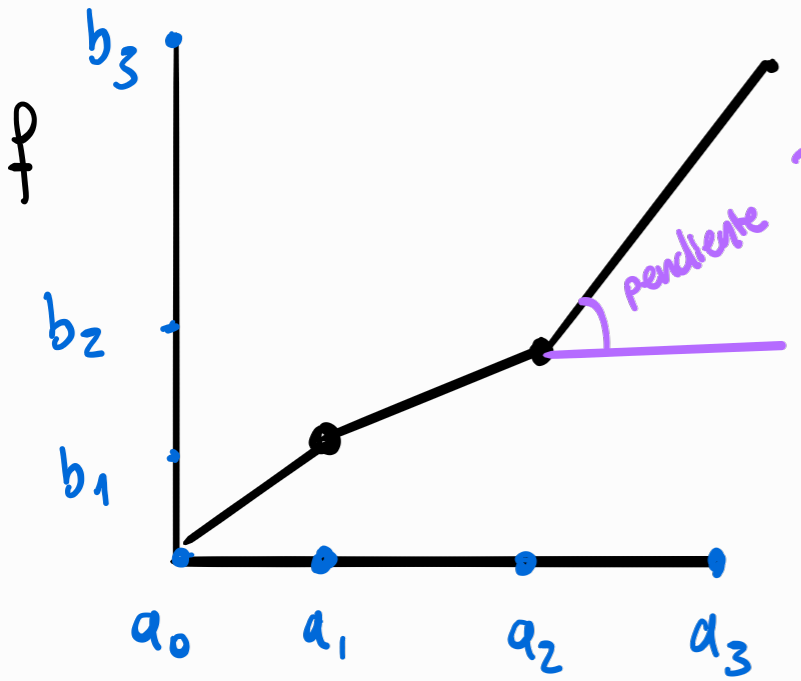




Pregunta $f \circ g$ se aplica primero de derecha a izq.

$f \circ g(x) = f(g(x))$.

Grupo de Thomson $F \leq \text{Homeo}_+[0,1]$.



- lineales por pedazos
- puntos de quiebre

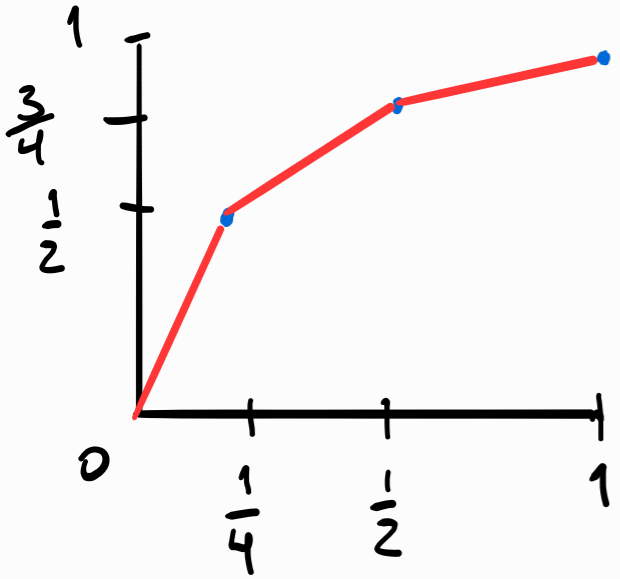
$$a_i, b_i \in \mathbb{Z}[1/2] \cap (0,1)$$

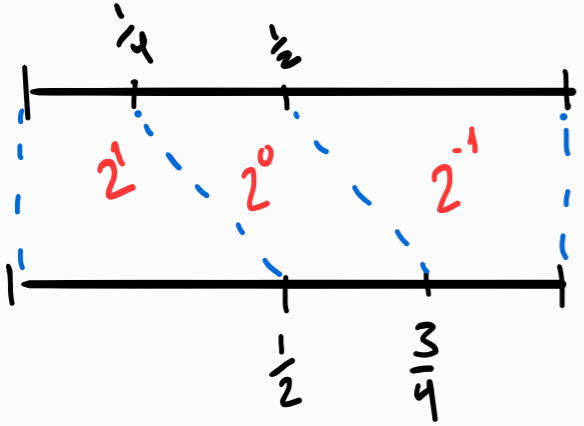
$$\| $$

$$\left\{ \frac{n}{2^m} \in \mathbb{Q} \;\middle|\; 0 < \frac{n}{2^m} < 1,\ n, m \in \mathbb{Z} \right\}$$

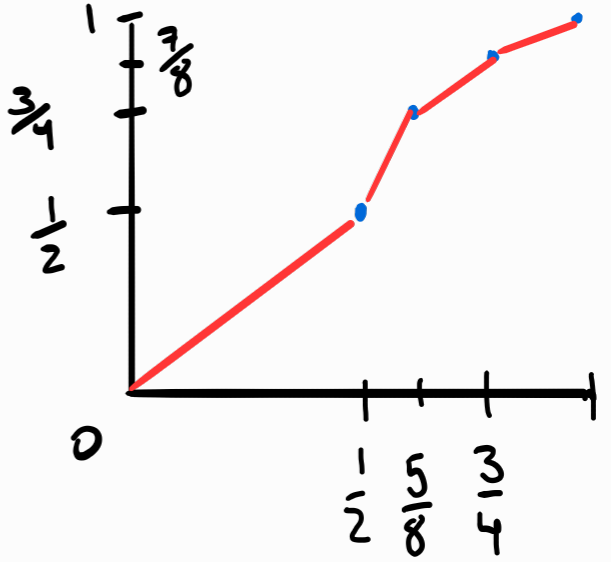
Cualquier $f \in F$ se determina por las subdivisiones en el eje horizontal y vertical.

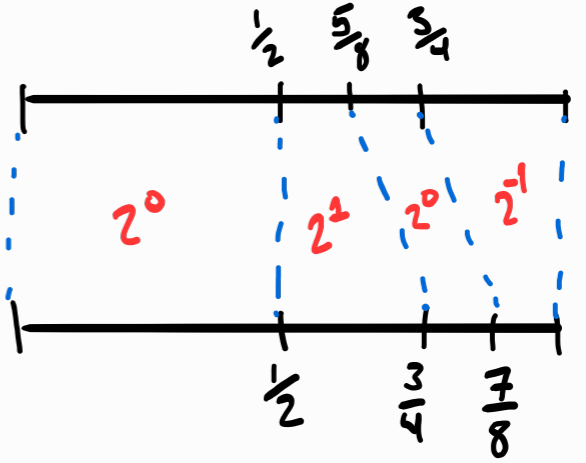
a:





b:





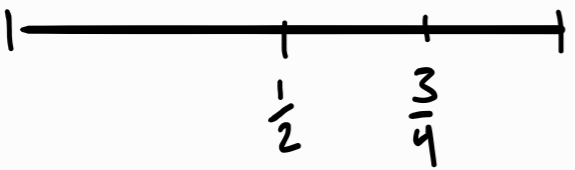
Codificación de subdivisiones

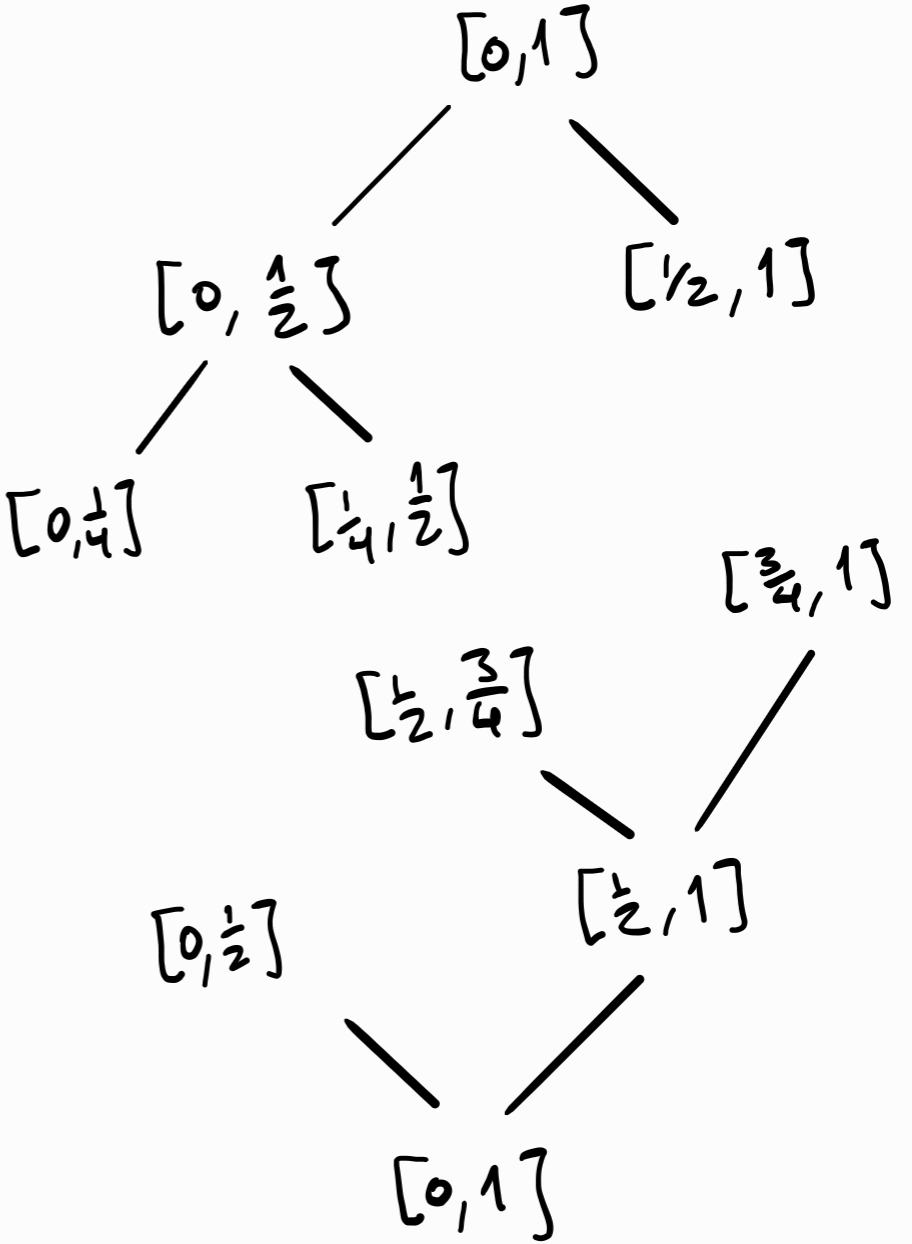
Árboles binarios finitos

Dominio



Codominio





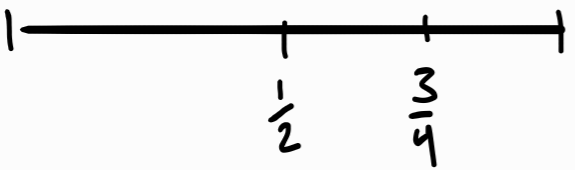
Codificación de subdivisiones

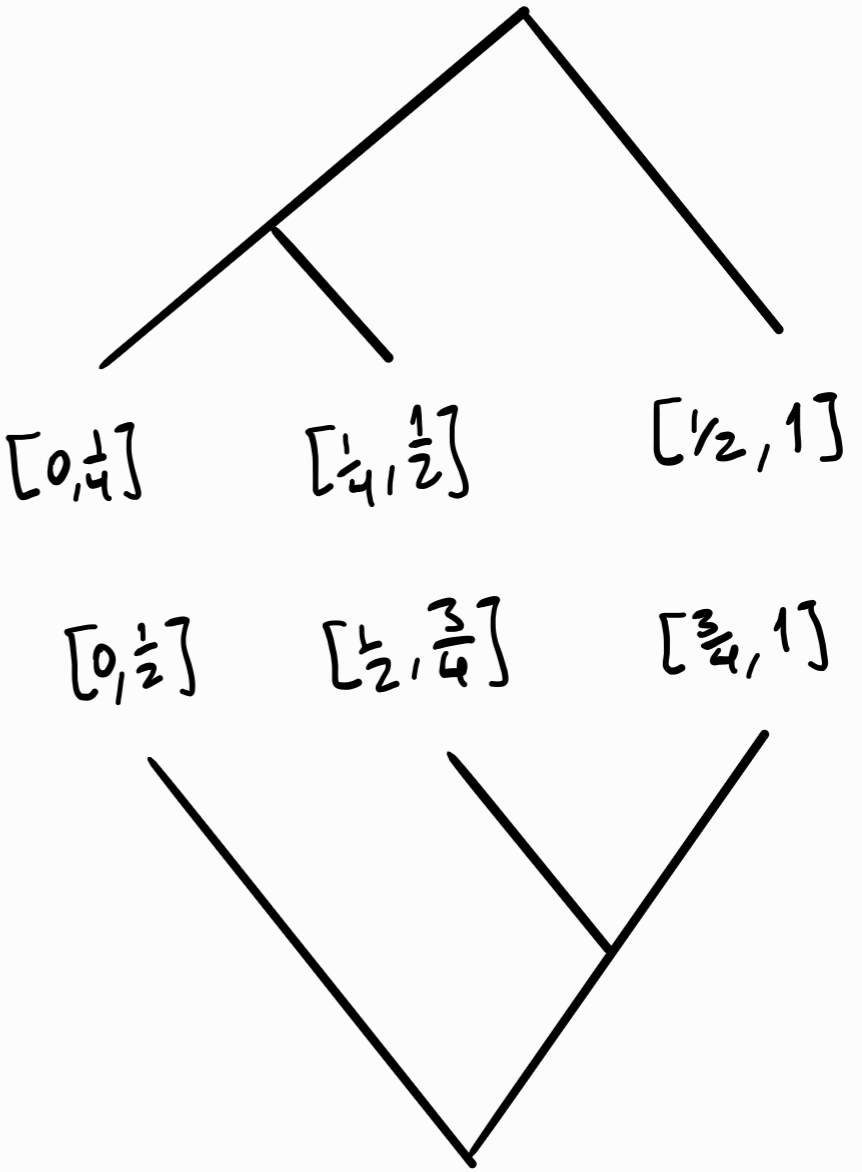
Árboles binarios finitos

Dominio



Codominio



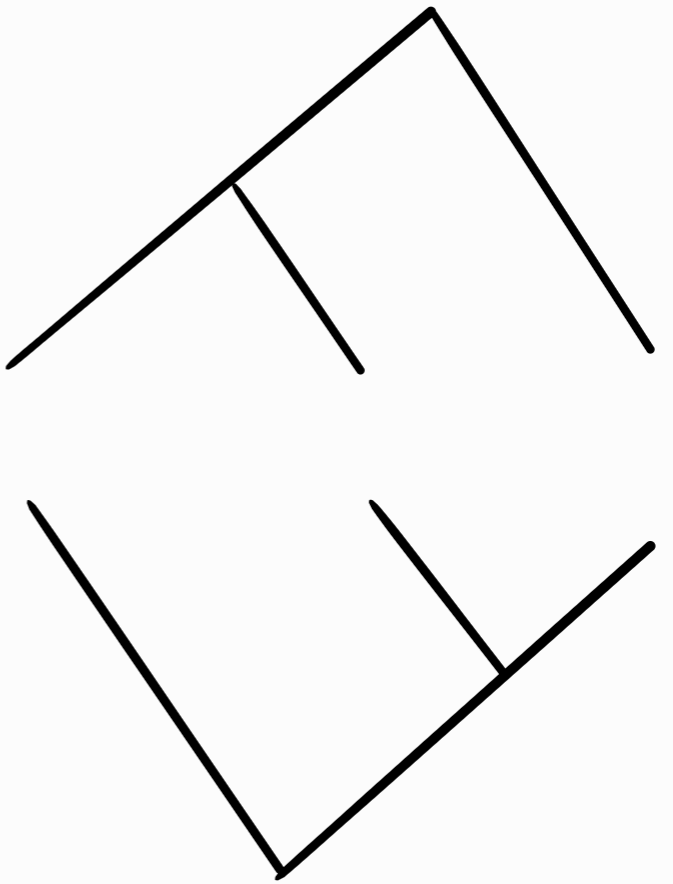


Codificación de subdivisiones

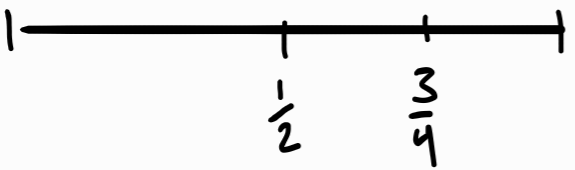
Árboles binarios finitos

Dominio



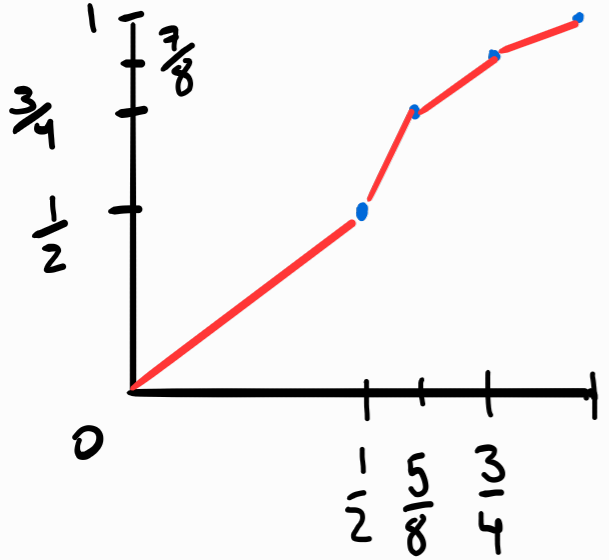


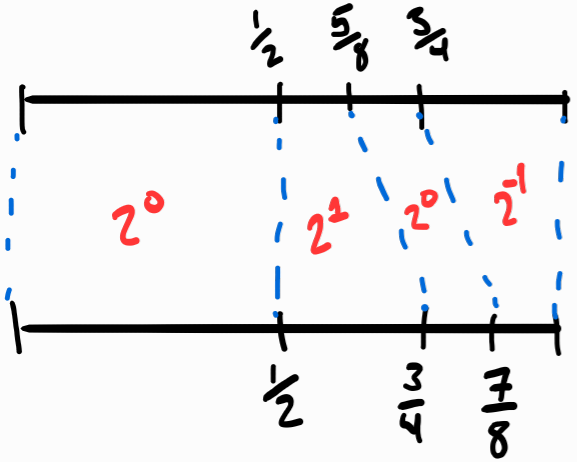
Codominio



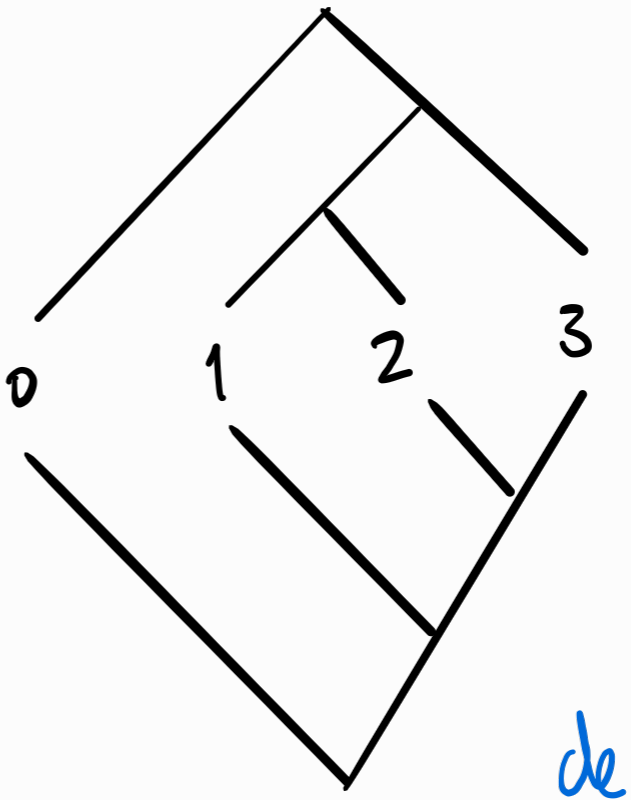
Similarmente b

b:





se representa como



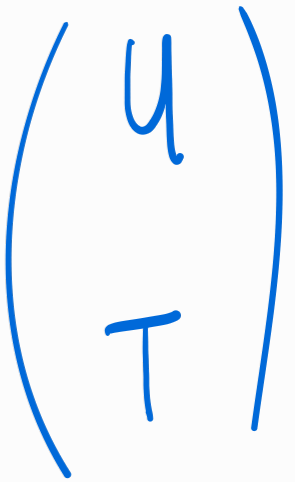
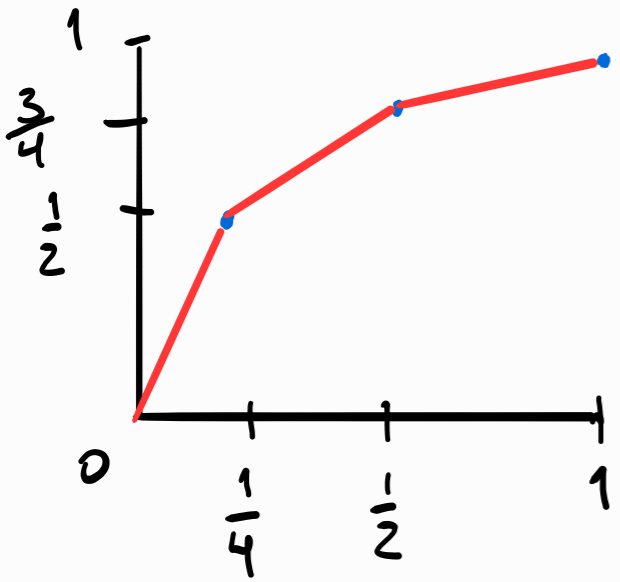


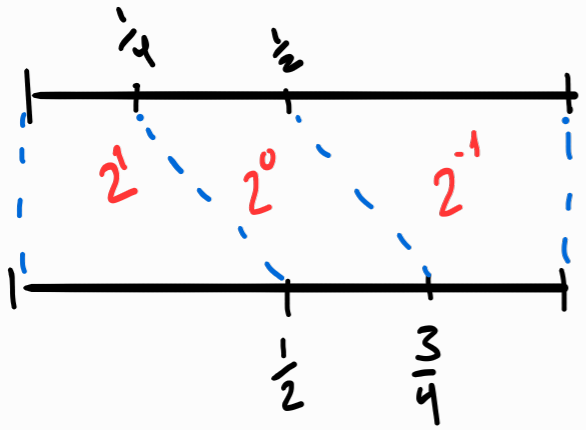
Diagrama de pares de árboles

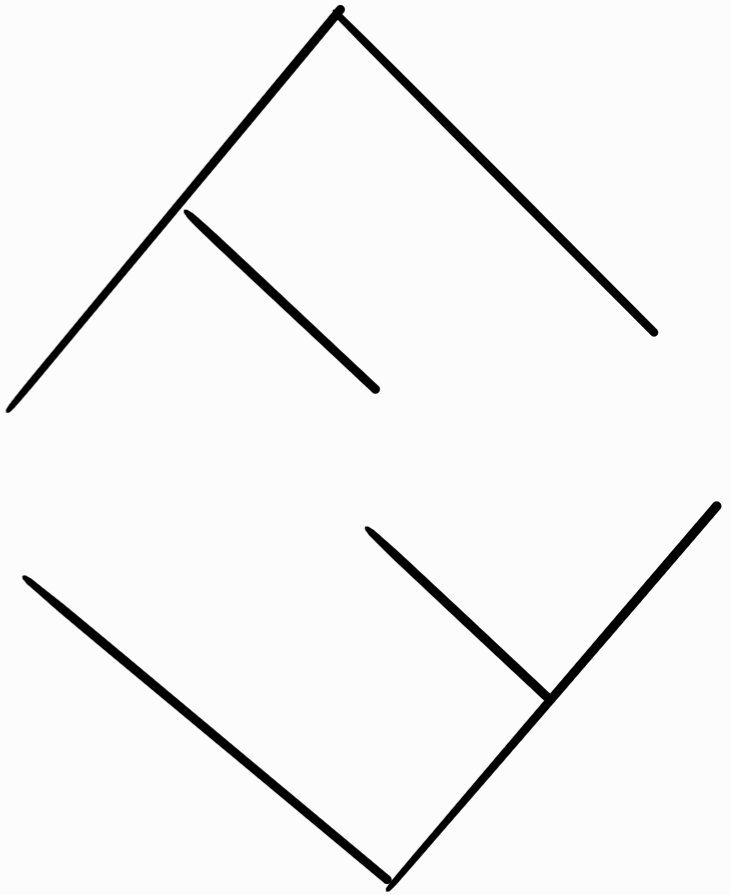
mismo # hojas

Más de un representante para un mismo homeomorfismo

a:

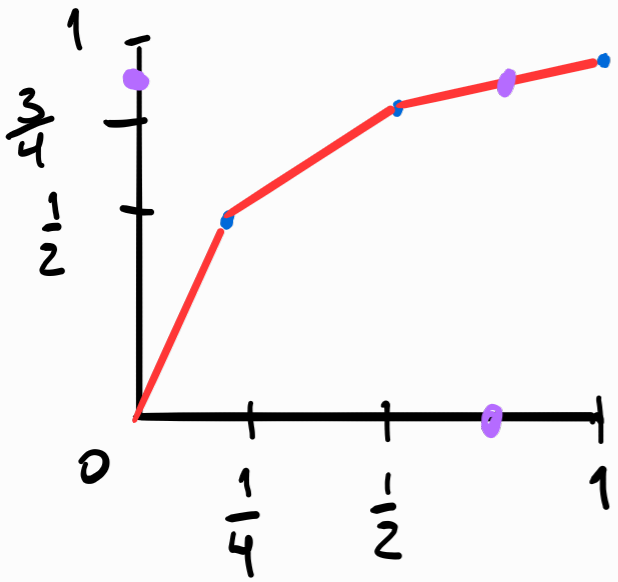


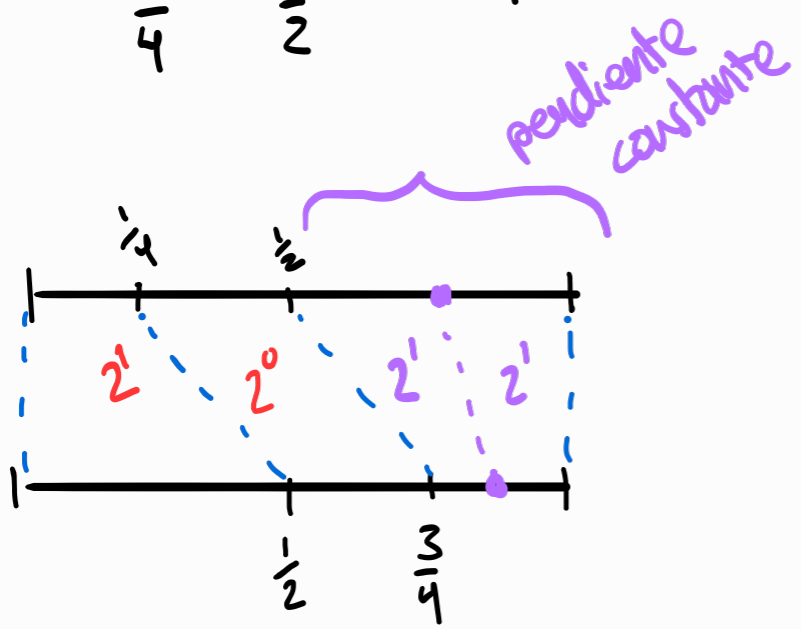


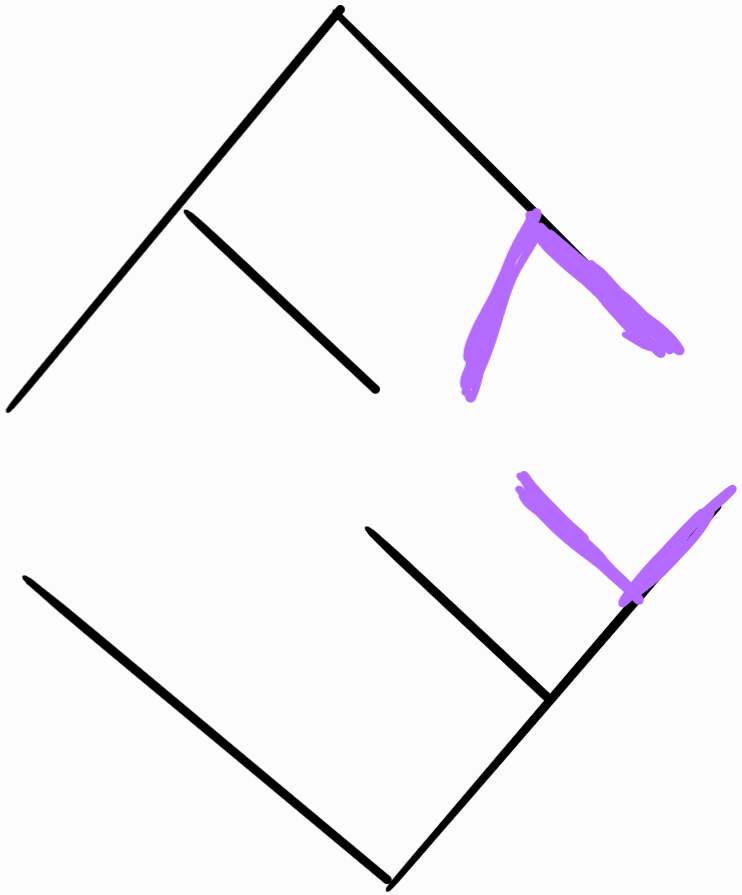


Más de un representante para un mismo homeomorfismo

a:

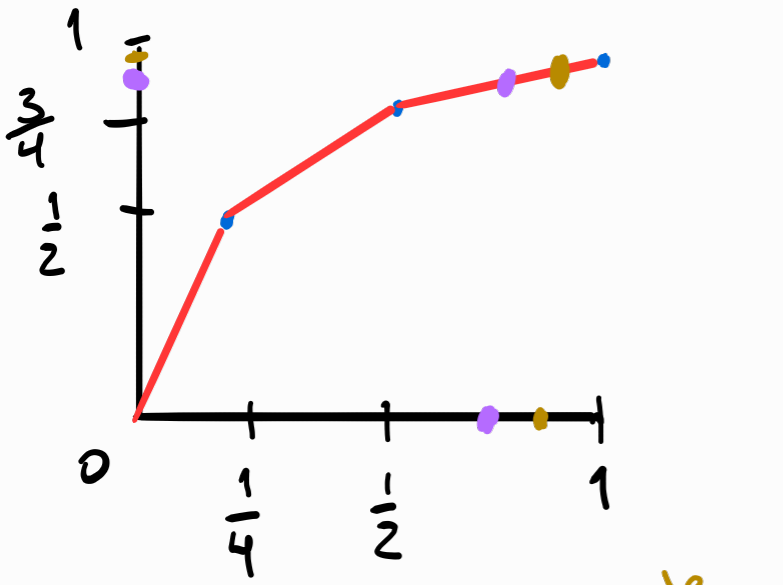


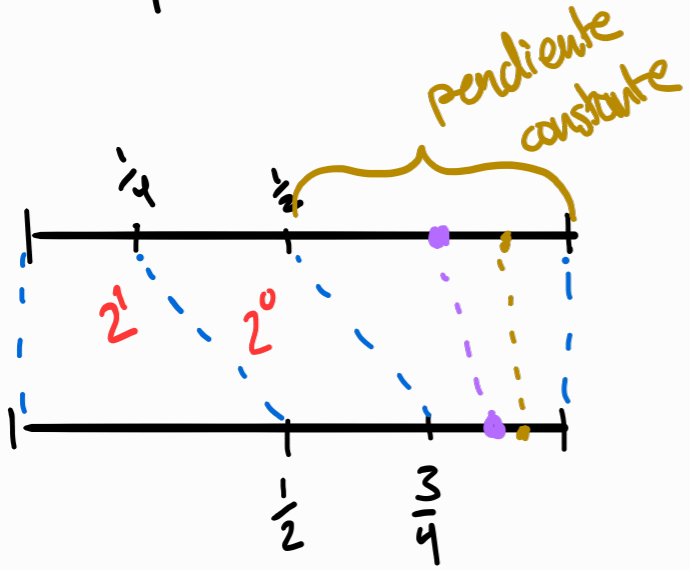


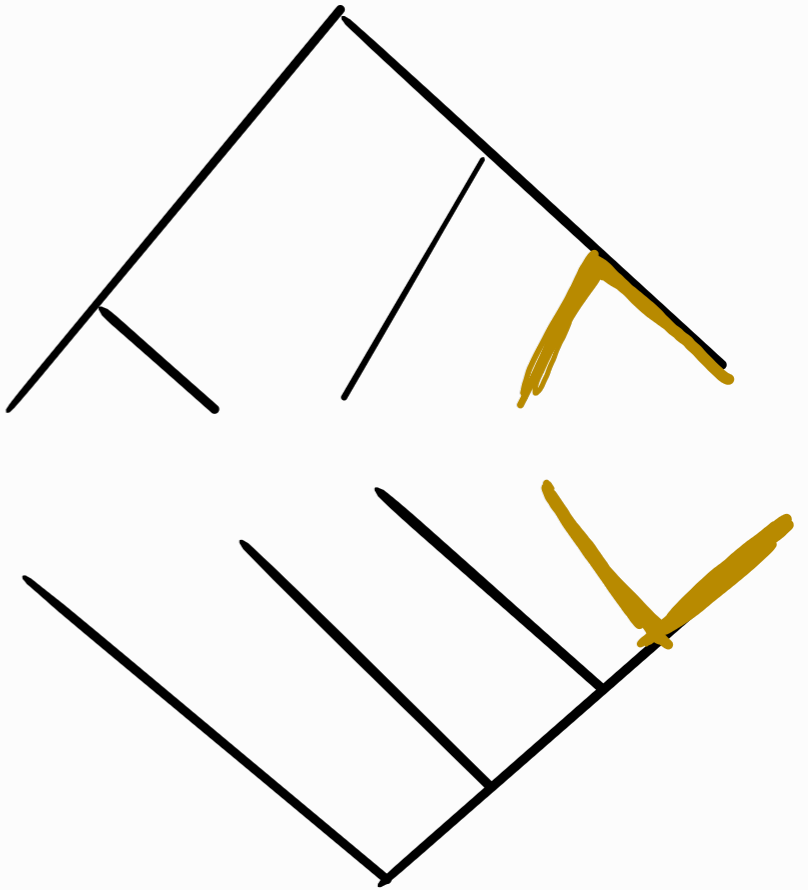


Más de un representante para un mismo homeomorfismo

a:

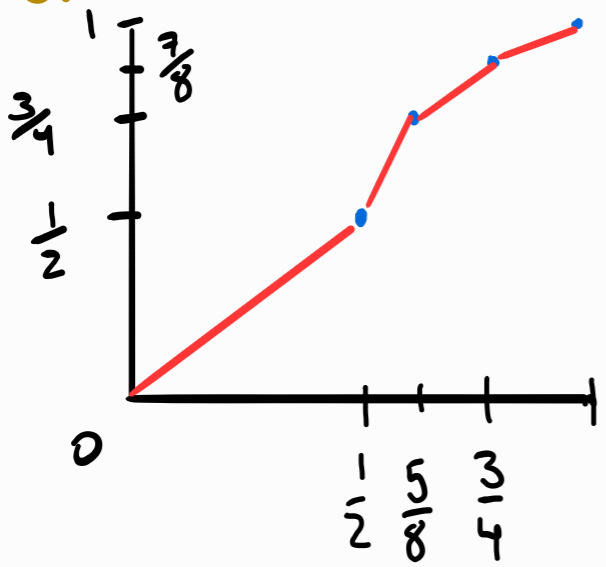


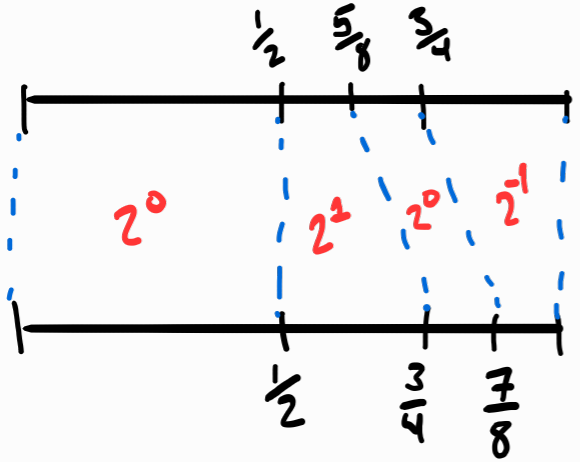


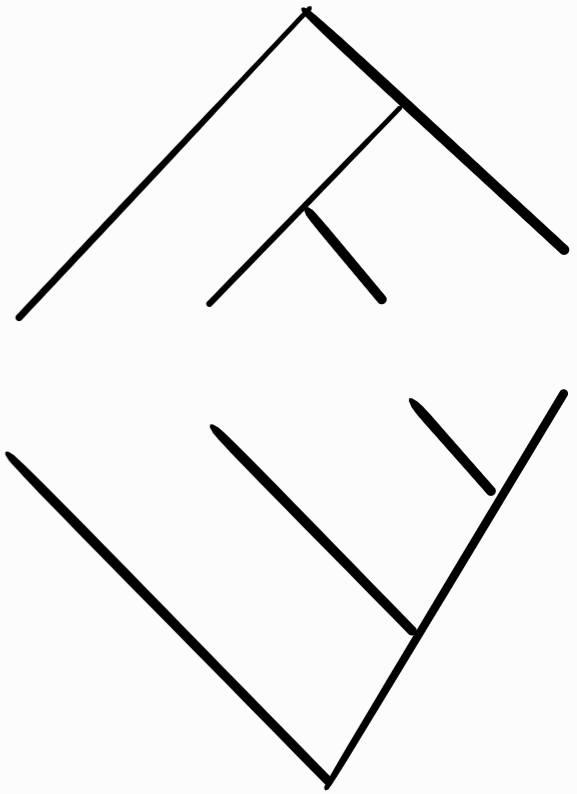


Similarmente b tiene otro representante.

b:

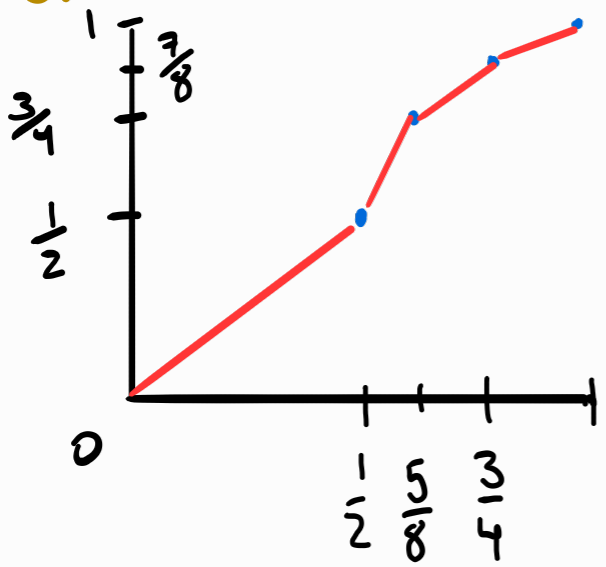


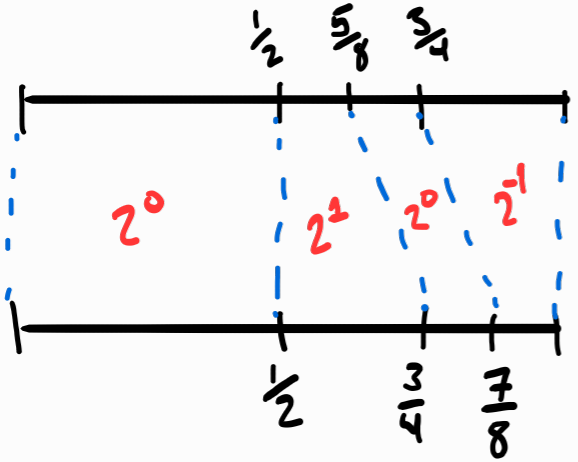


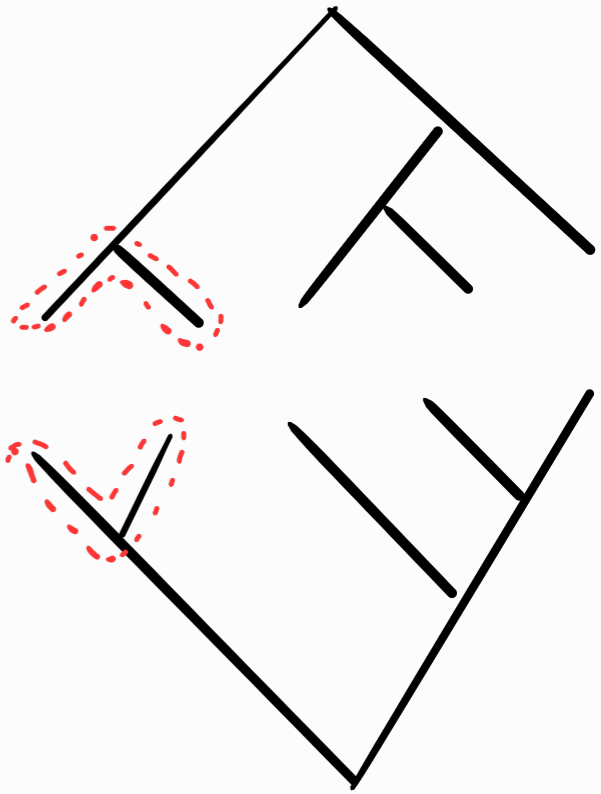


Similarmente b tiene otro representante.

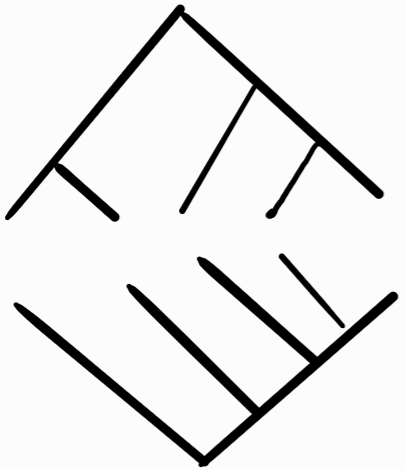
b:

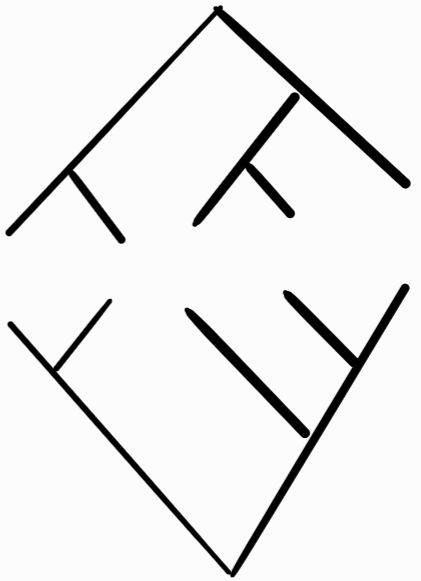


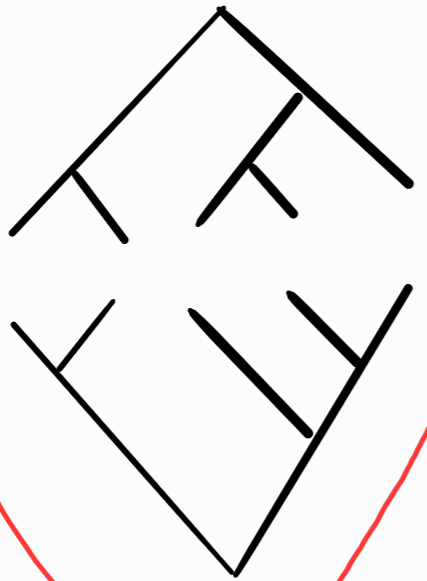
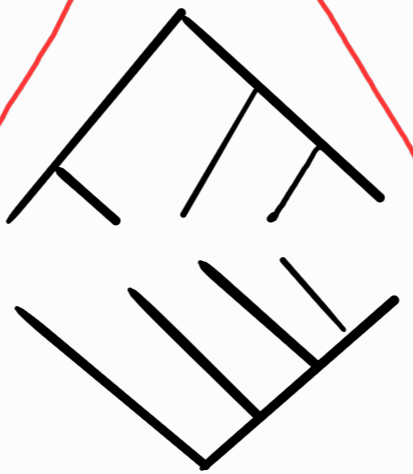
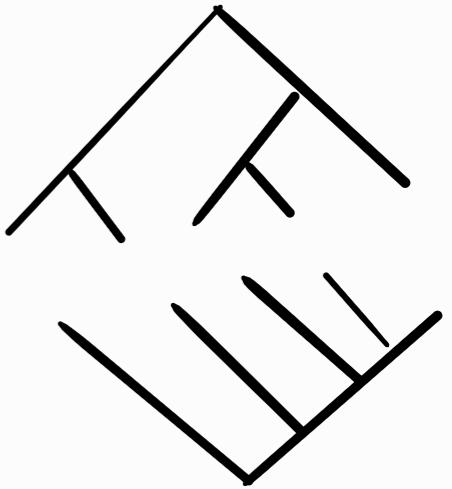




Cambiar de representantes nos ayuda a operar

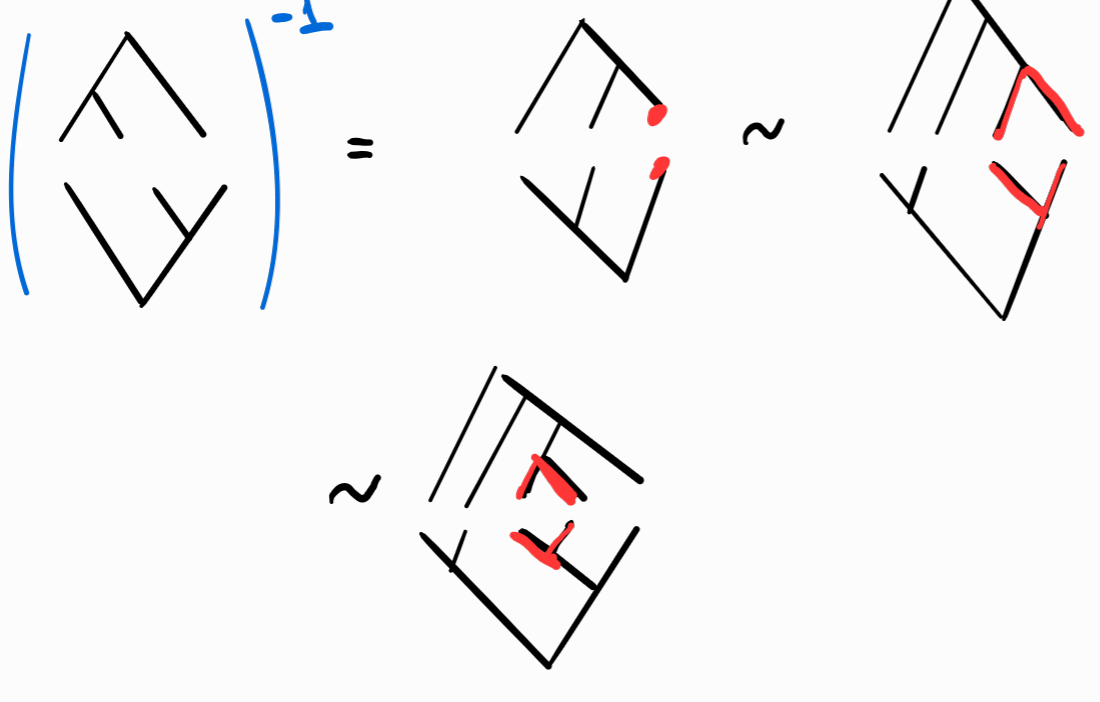
a:  $\begin{pmatrix} u \\ s \end{pmatrix}$

b:  $\begin{pmatrix} T \\ u \end{pmatrix}$

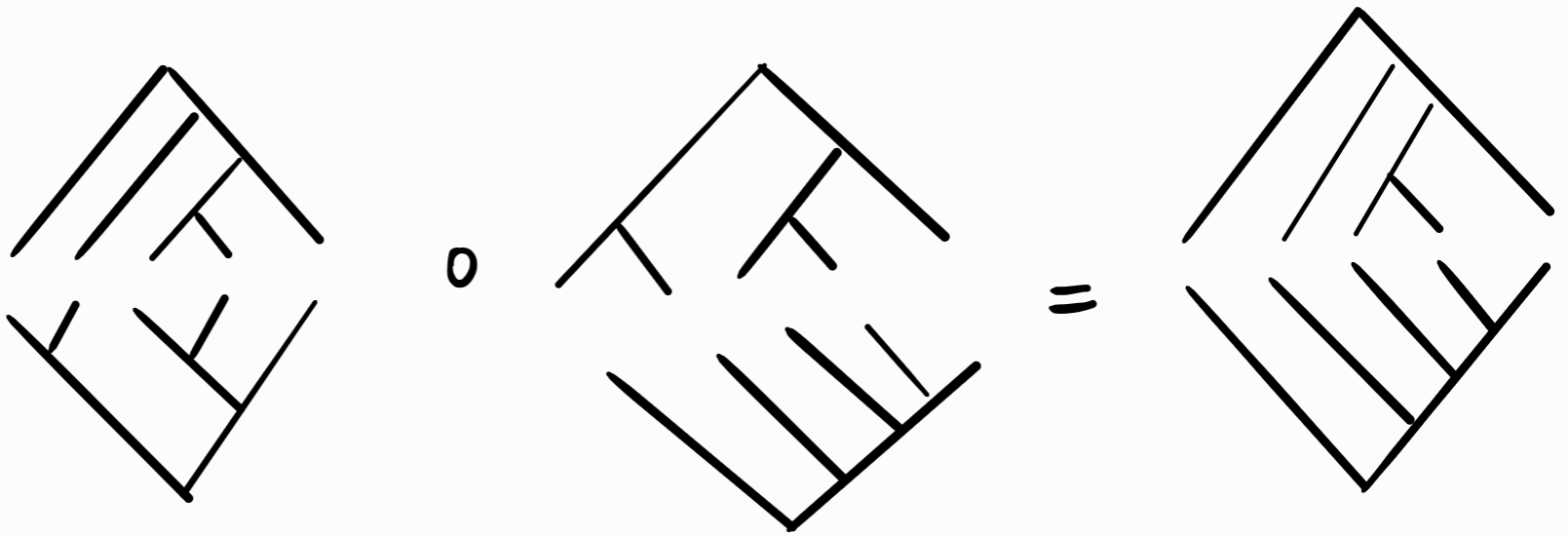
ba:  ∘  = 

¿No les suena a que en realidad calculamos $a^{-1} b^{-1}$?

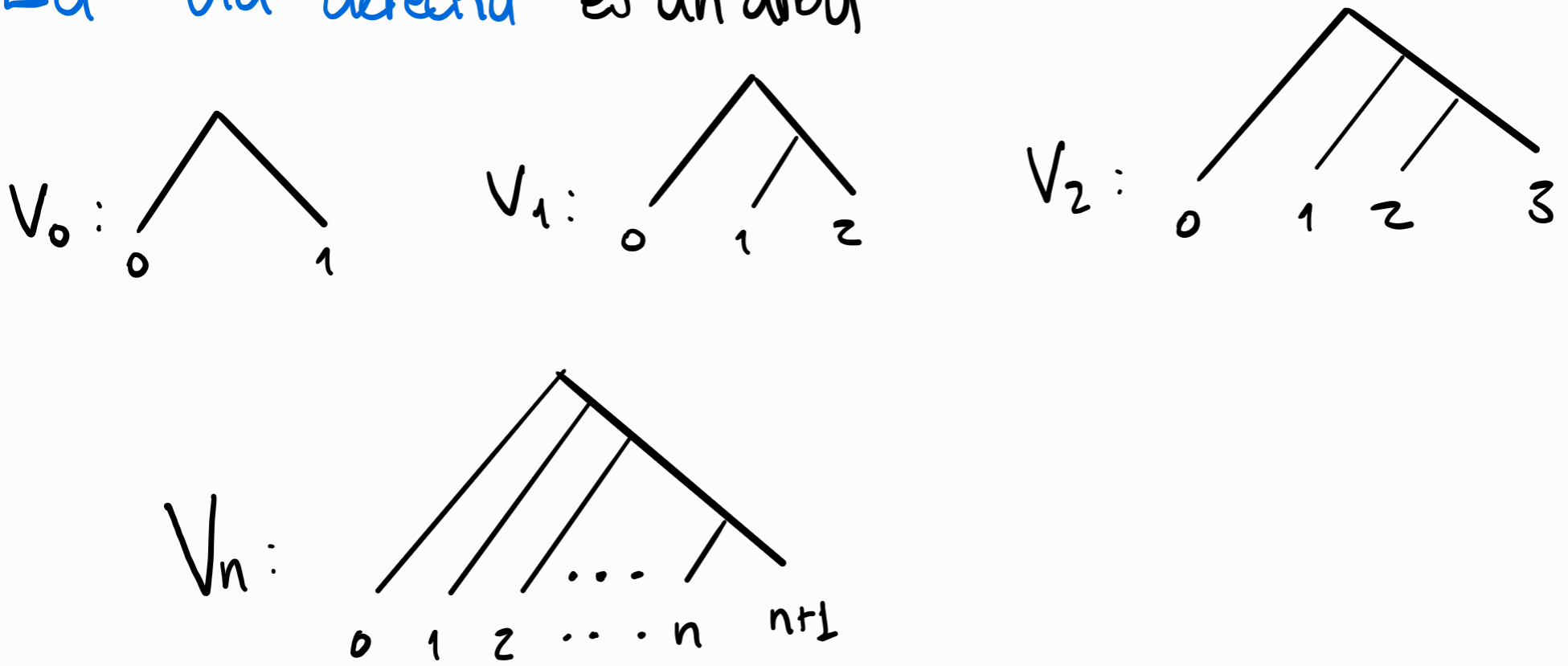
a^{-1} :



$a^{-1}ba$:



La vid derecha es un árbol



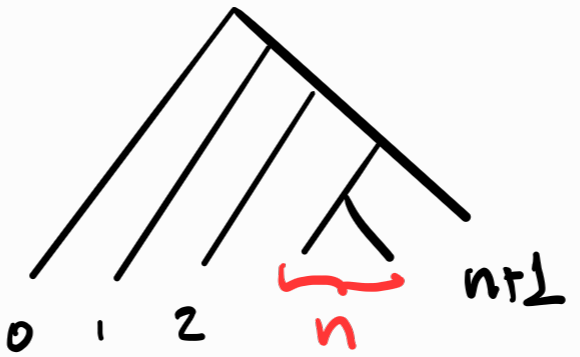
Elemento positivo es $\begin{bmatrix} T \\ V_n \end{bmatrix}$ T árbol binario de n+2 hojas

$\vdots$

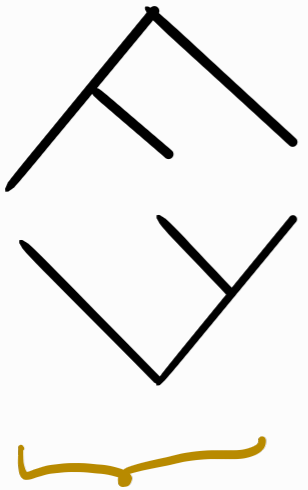
denotado por $[T]$

Elementos positivos más básicos $x_n = [V_n \wedge n]$

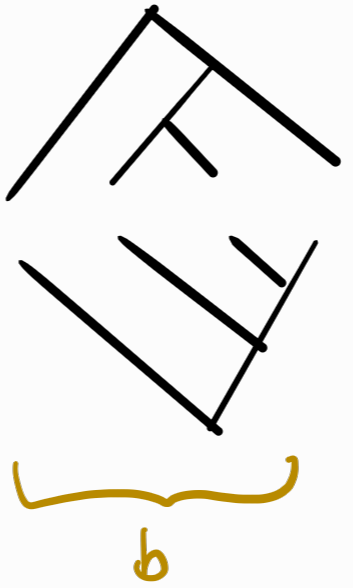
$V_n \wedge n$:



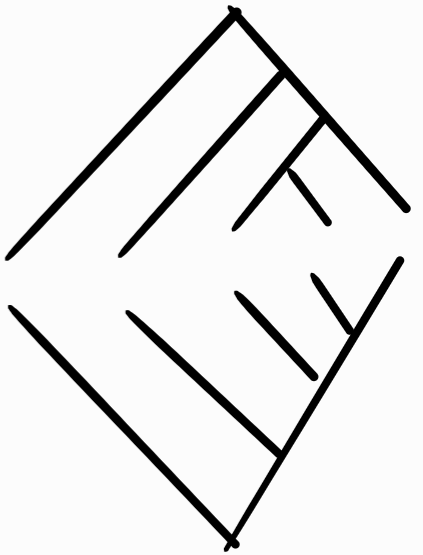
$x_0 =$



$x_1 =$



x_2 :



Exposición 1 Demostrar que $x_i^{-1} x_j x_i = x_{j+1}$ $\forall i < j$.

Hint: $i \geq n \Rightarrow x_n = [V_n \wedge n] = [V_i \wedge n]$

Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$.

Hint: $i \geq n \Rightarrow x_n = [V_n \wedge n] = [V_i \wedge n]$

Inducción. $i = n$ es cierto.

Sup. cierto para $i \geq n$.

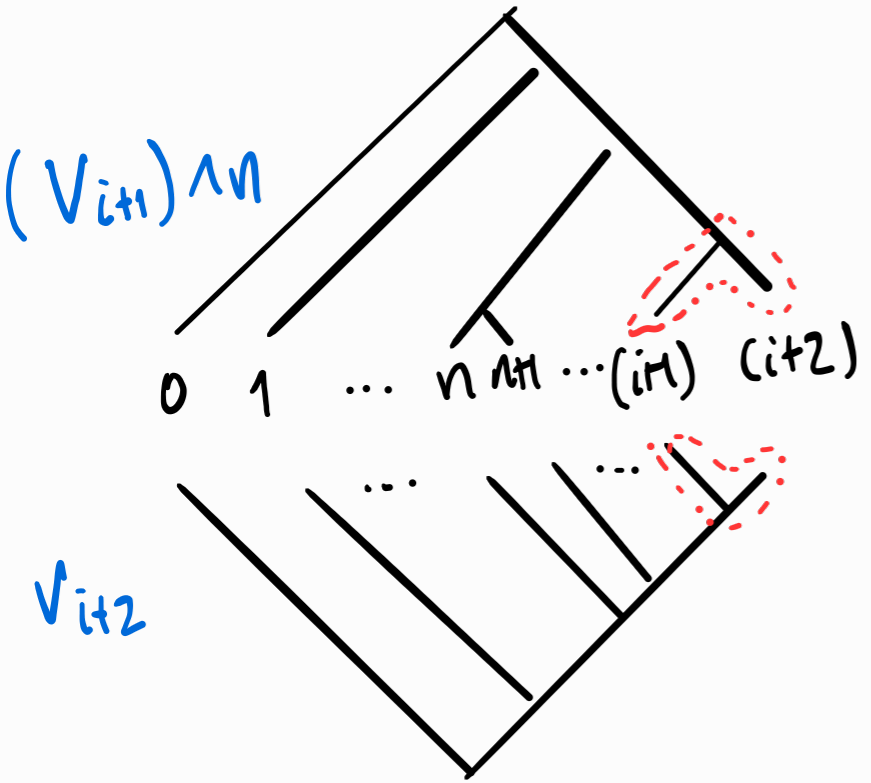
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$.

Hint: $i \geq n \Rightarrow x_n = [V_n \wedge n] = [V_i \wedge n]$

Inducción. $i = n$ es cierto.

Sup. cierto para $i \geq n$.



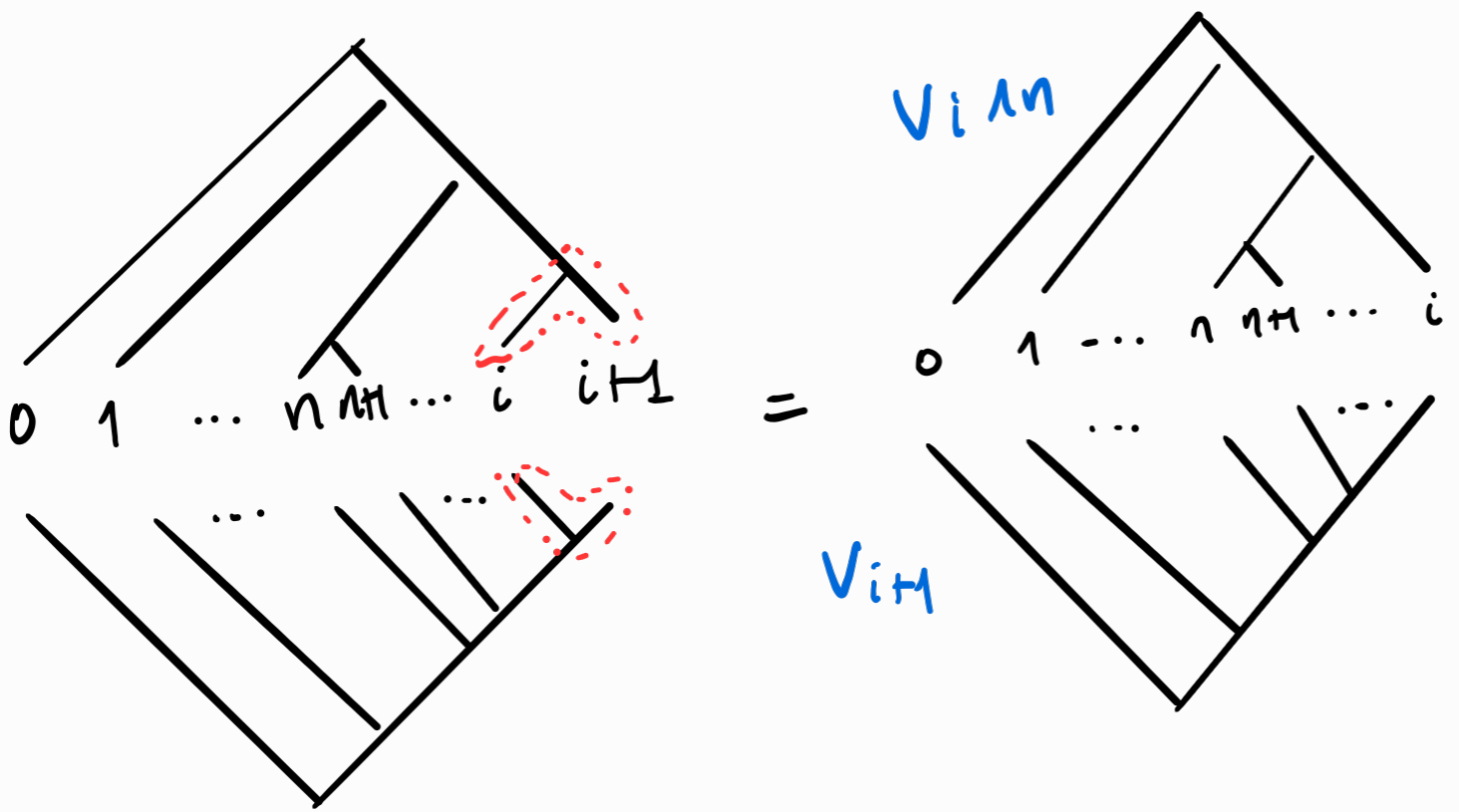
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$.

Hint: $i \geq n \Rightarrow x_n = [V_n \wedge n] = [V_i \wedge n]$

Inducción. $i = n$ es cierto.

Sup. cierto para $i \geq n$.



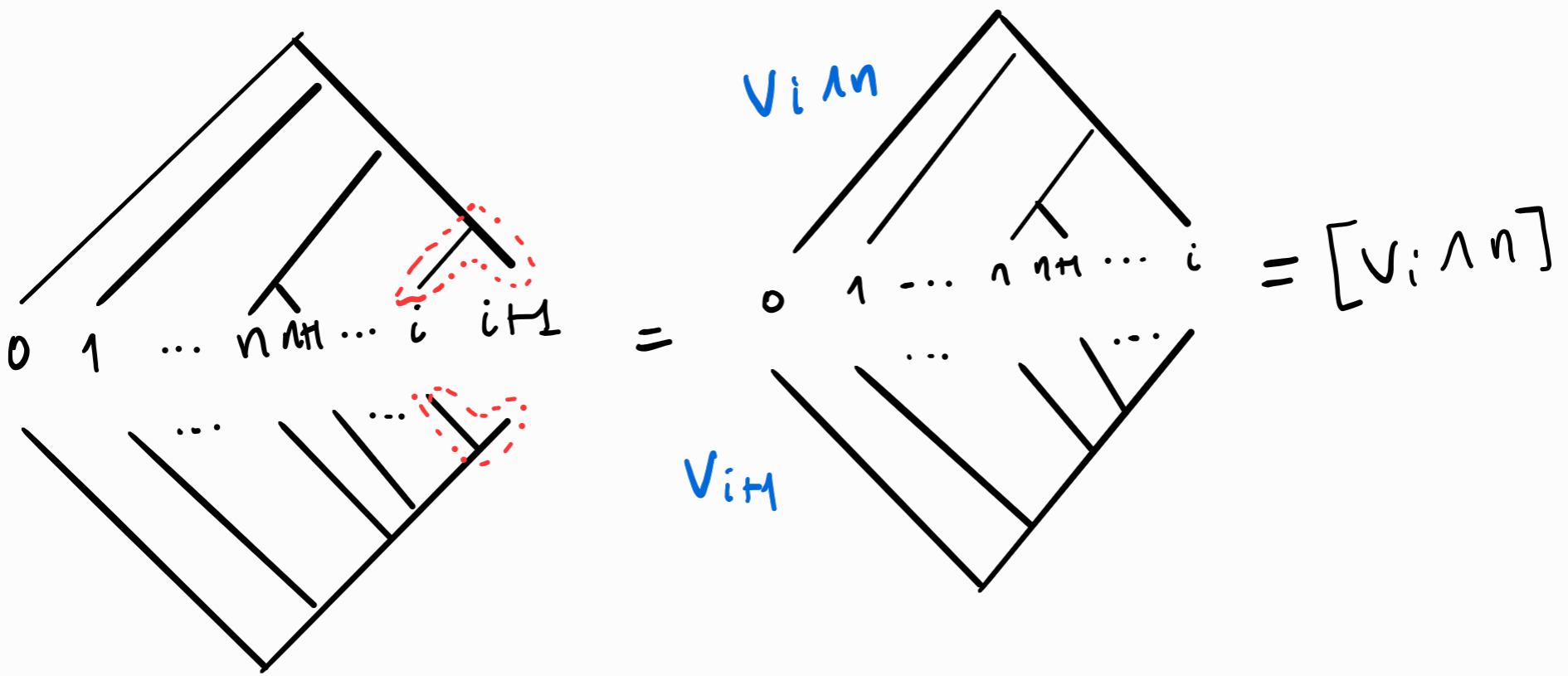
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$.

Hint: $i \geq n \Rightarrow x_n = [V_n \wedge n] = [V_i \wedge n]$

Inducción. $i = n$ es cierto.

Sup. cierto para $i \geq n$.



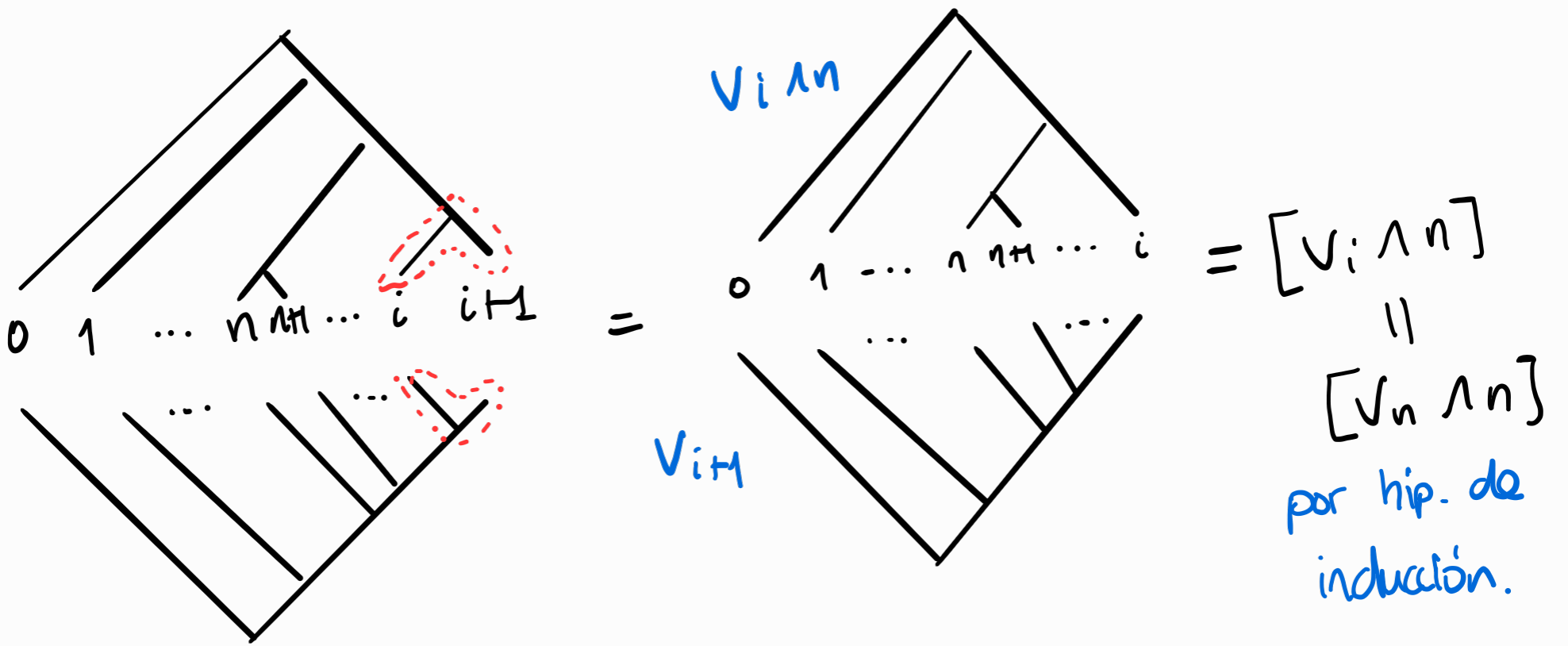
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$.

Hint: $i \geq n \Rightarrow x_n = [V_n \wedge n] = [V_i \wedge n]$

Inducción. $i = n$ es cierto.

Sup. cierto para $i \geq n$.



Hint 2. Si T tiene al menos $n+2$ hojas $[T \wedge n] = [T] \cdot x_n$.

$k+2 = \#$ hojas $T \Rightarrow k \geq n$.

$$[T] \cdot x_n = [T] \cdot [V_k \wedge n]$$

Hint 2. Si T tiene al menos $n+2$ hojas $[T \wedge n] = [T] \cdot x_n$.

$k+2 = \#$ hojas $T \Rightarrow k \geq n$.

$$[T] \cdot x_n = [T] \cdot [V_k \wedge n] = \begin{bmatrix} T \\ V_k \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

Hint 2. Si T tiene al menos $n+2$ hojas $[T \wedge n] = [T] \cdot x_n$.

$k+2 = \#$ hojas $T \Rightarrow k \geq n$.

$$[T] \cdot x_n = [T] \cdot [V_k \wedge n] = \begin{bmatrix} T \\ V_k \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

$$= \begin{bmatrix} T \wedge n \\ V_k \wedge n \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

Hint 2. Si T tiene al menos $n+2$ hojas $[T \wedge n] = [T] \cdot x_n$.

$k+2 = \#$ hojas $T \Rightarrow k \geq n$.

$$[T] \cdot x_n = [T] \cdot [V_k \wedge n] = \begin{bmatrix} T \\ V_k \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

$$= \begin{bmatrix} T \wedge n \\ V_k \wedge n \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

$$= \begin{bmatrix} T \wedge n \\ V_{k+1} \end{bmatrix}$$

Hint 2. Si T tiene al menos $n+2$ hojas $[T \wedge n] = [T] \cdot x_n$.

$k+2 = \#$ hojas $T \Rightarrow k \geq n$.

$$[T] \cdot x_n = [T] \cdot [V_k \wedge n] = \begin{bmatrix} T \\ V_k \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

$$= \begin{bmatrix} T \wedge n \\ V_k \wedge n \end{bmatrix} \cdot \begin{bmatrix} V_k \wedge n \\ V_{k+1} \end{bmatrix}$$

$$= \begin{bmatrix} T \wedge n \\ V_{k+1} \end{bmatrix} = [T \wedge n].$$

Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i] \quad \text{por H2}$$

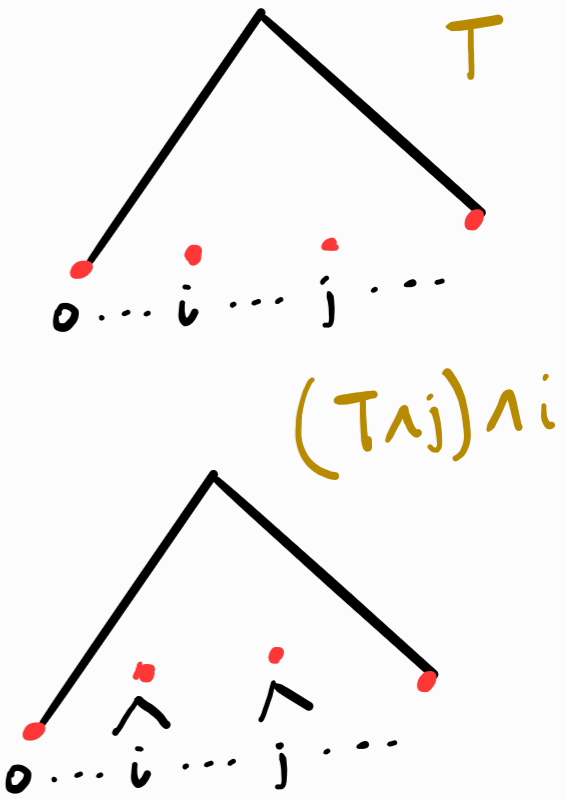
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i \wedge (j+1)]$$



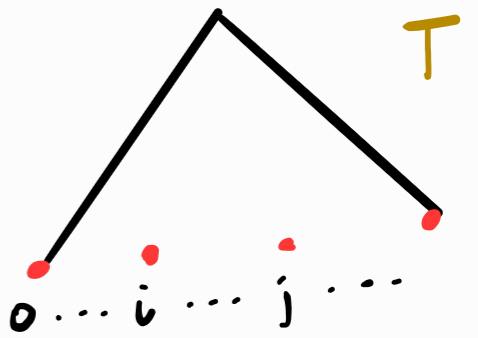
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

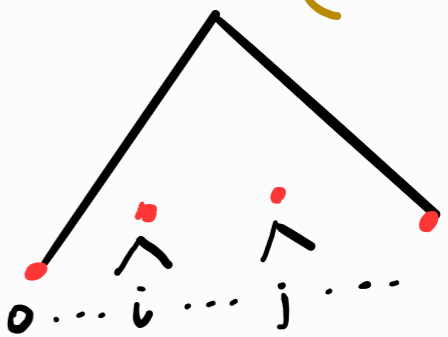
$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i]$$

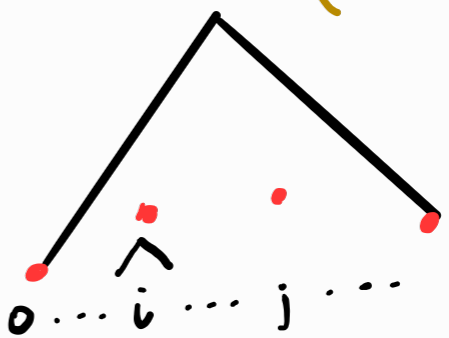
$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i \wedge (j+1)]$$



$(T \wedge j) \wedge i$



$(T \wedge i)$



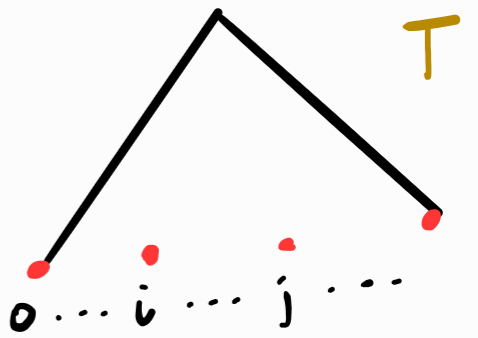
Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

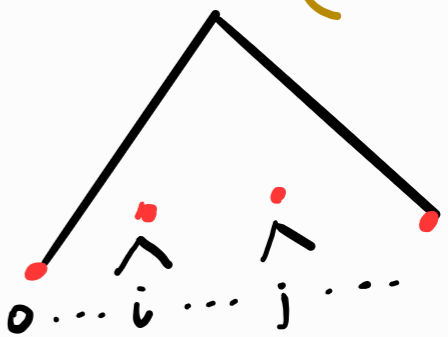
$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i]$$

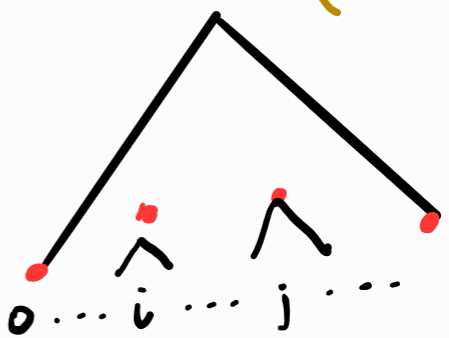
$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i \wedge (j+1)]$$



$(T \wedge j) \wedge i$



$(T \wedge i) \wedge (j+1)$



Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i \wedge (j+1)]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i] \cdot x_{j+1}$$

Por H2

Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i \wedge (j+1)]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i] \cdot x_{j+1}$$

$$= [v_i \wedge i]^{-1} \cdot [v_i \wedge i] \cdot x_{j+1}$$

Por H1

Exposición 1

Demostrar que $x_i^{-1} x_j x_i = x_{j+1} \quad \forall i < j$

$$x_i^{-1} x_j x_i = [v_i \wedge i]^{-1} \cdot [v_j \wedge j] \cdot x_i$$

$$= [v_i \wedge i]^{-1} \cdot [(v_j \wedge j) \wedge i]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i \wedge (j+1)]$$

$$= [v_i \wedge i]^{-1} \cdot [v_j \wedge i] \cdot x_{j+1}$$

$$= [v_i \wedge i]^{-1} \cdot [v_i \wedge i] \cdot x_{j+1}$$

$$= x_{j+1}$$